

Arthur Sumague

Data & Analytics Developer

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Data engineer with a graduate background in data science and production experience building Databricks / Delta Lake pipelines, dbt transformations, and BI-ready data models in the energy sector. Strong in Python, SQL, and PySpark; owns ETL/ELT, dimensional modeling, incremental / SCD patterns, and data-quality testing end to end.

SKILLS

Languages: Python, SQL, T-SQL, PySpark, R, TypeScript, DAX, M (Power Query)

Data Engineering & ETL: Databricks, Apache Spark, Delta Lake, dbt, Apache Airflow, SQL Server / SSIS, DuckDB, PostgreSQL, data warehousing, star schema, SCD, CDC, incremental loads

Cloud: Azure (Databricks, Synapse, ADLS, Key Vault), AWS (EC2, S3, RDS, Athena)

BI & Reporting: Power BI, Power Query, DAX, Tabular Editor, Power Apps

ML & Data Science: pandas, NumPy, scikit-learn, statsmodels, MLflow, XGBoost, PyTorch, Plotly

Practices & Tooling: Git, CI/CD, REST APIs, data-quality testing, Linux, uv, Unity Catalog governance

EXPERIENCE

Data Analytics Developer I (promoted from Intern) — Dakota Analytics *Apr 2025 – Present · Calgary, AB*

- Built an incremental change-detection and history-archiving pipeline on **Databricks (PySpark, Delta Lake, Unity Catalog)** that snapshots amended emissions records into versioned amendment chains for Alberta regulatory compliance (AMD), ingesting from SQL Server 2016 via JDBC.
- Designed a **Databricks-to-Power BI reporting architecture** that pushes transformation logic into dbt / Spark SQL with materialized hashed surrogate keys and a thin DAX layer, improving testability and refresh reliability.
- Scoped and assessed **2,000+ legacy reports and 20+ source systems** for a major O&G producer to rationalize the reporting estate and guide a modern data-platform migration.
- Developed a **data-quality integrity platform** validating record sync across 3 databases along the pipeline, catching drift before it reached reporting.
- Developed production & sales reporting for an O&G explorer/producer surfacing daily volumes per well; modeled and integrated **5 upstream production data sources** via ETL.
- Built a robotic-process-automation workflow over an in-house file directory, identified to automate **~900 hours** of annual engineering effort.

Undergraduate Researcher — APACE Laboratory *May 2022 – Aug 2023*

- protein-structure methodology development (HDX-MS), recombinant protease expression.
- practice of scientific method to develop scalable production of proteases.

SELECTED PROJECTS

- **Energy Analytics & Drilling Forecast Platform** — End-to-end ML pipeline over 26 years / 294K+ EIA records; medallion-architecture PostgreSQL warehouse (bronze→silver→gold) with OOP ingestion classes and dbt Core transforms; SARIMAX rig-count forecasting with ADF/Granger feature selection and MLflow tracking. *(Python, dbt, PostgreSQL, statsmodels, pmdarima, MLflow)*
- **FX Ticker Predictor** — Deployed 3-tier AWS pipeline (EC2/S3) forecasting daily FX High/Low bands for EUR/GBP/JPY; fuses finBERT news-sentiment with OHLC data into a Prophet model; idempotent incremental ETL with a Streamlit dashboard. *(Python, AWS, finBERT/PyTorch, Prophet, Selenium)*
- **SkillsSyncRAG** — RAG job-market intelligence tool; ingests live postings into a ChromaDB vector store and orchestrates retrieval + GPT-4o skill ranking with a LangChain ReAct agent behind a Streamlit UI. *(Python, LangChain, ChromaDB, OpenAI)*

EDUCATION

M.S., Data Science & Analytics — University of Calgary *2025*

B.Sc. (Honours), Biochemistry — University of Calgary *2024*